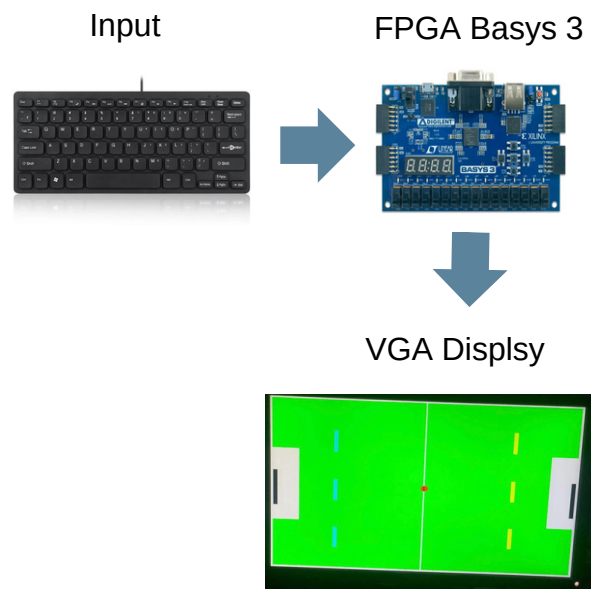
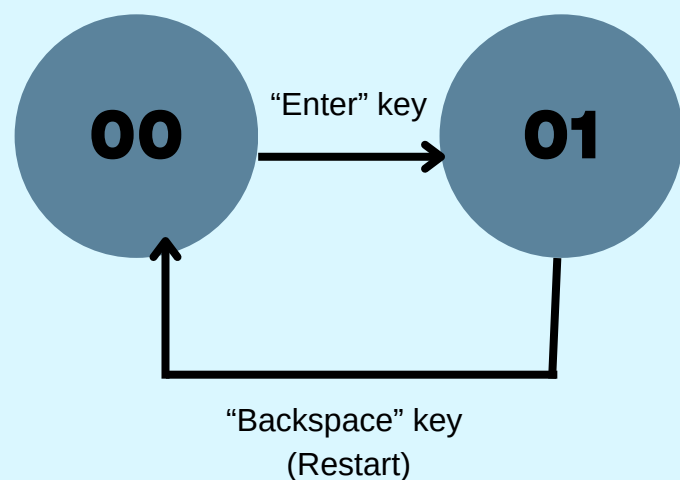


System diagram



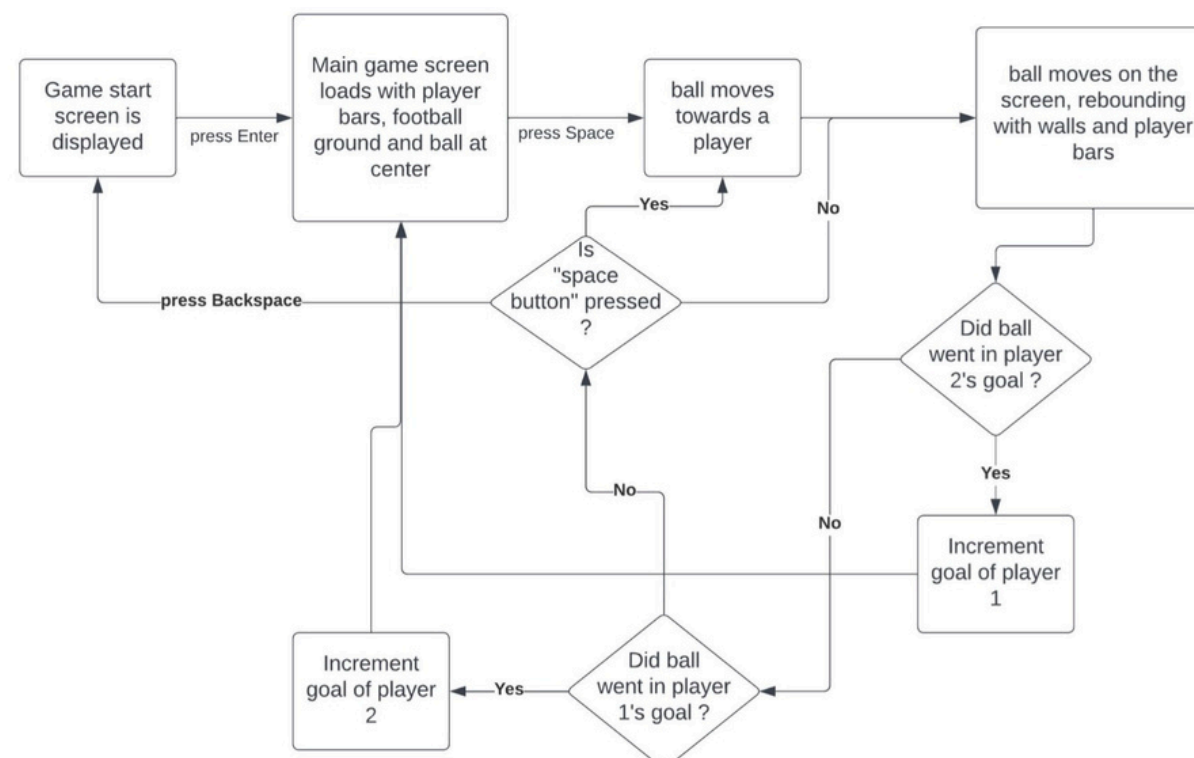
FSM state transition Diagram



INTRODUCTION

With our mission to develop a captivating foosball game that maximizes the capabilities of the Basys 3 board, our project features a two player game that puts players in control of two bars each, maneuvering them with precision using keyboard inputs. With interactive features like a quick-reset button for seamless restarts and a screen toggle option for added convenience, players can stay focused on the excitement of the game. Each round begins with anticipation as players press the enter key on the start screen, and the competitive energy ramps up with the spacebar hit that kicks off gameplay. We also have a score counter that displays the points scored by each player. Our game is based on the execution of a Melay machine in which the inputs are considered as a factor in determining the next state.

User flow diagram



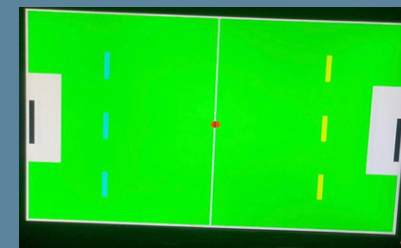
STATE 00

This is the start screen. The transition from the start screen to the main game screen is initiated by pressing the "Enter" key.



STATE 01

This is the main game screen. Pressing the "Backspace" key will reset the game and take the user back to the start screen



Hardware resource utilisation

